

Surgical outcome in mesial temporal sclerosis correlates with prion protein gene variant

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Abstract—Background: Mesial temporal lobe epilepsy related to hippocampal sclerosis (MTLE-HS) is the most common surgically remediable epileptic syndrome. Ablation of the cellular prion protein (PrP^c) gene (*PRNP*) enhances neuronal excitability of the hippocampus in vitro and sensitivity to seizure in vivo, indicating that PrP^c might be related to epilepsy. **Objective:** To evaluate the genetic contribution of *PRNP* to MTLE-HS. **Methods:** The *PRNP* coding sequence of DNA from peripheral blood cells of 100 consecutive patients with surgically treated MTLE-HS was compared to that from a group of healthy controls adjusted for sex, age, and ethnicity (n = 180). The presence of *PRNP* variant alleles was correlated with clinical and presurgical parameters as well as surgical outcome. **Results:** A variant allele at position 171 (Asn→Ser), absent in controls, was found in heterozygosis (Asn171Ser) in 23% of patients ($p < 0.0001$). The *PRNP* genotypes were not correlated with any clinical or presurgical data investigated. However, patients carrying the Asn171Ser variant had a five times higher chance of continuing to have seizures after temporal lobectomy (95% CI 1.65 to 17.33, $p = 0.005$) than those carrying the normal allele. At 18 months after surgery, 91.8% of patients with the normal allele at codon 171 were seizure free, in comparison to 68.2% of those carrying Asn171Ser ($p = 0.005$). **Conclusions:** The *PRNP* variant allele Asn171Ser is highly prevalent in patients with medically untreatable MTLE-HS and influences their surgical outcome. The results suggest that the *PRNP* variant allele at codon 171 (Asn171Ser) is associated with epileptogenesis in MTLE-HS.

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Epilepsy is a health problem with a prevalence of 5 to 10 cases per 1,000 habitants in developed countries, which may be even higher in developing countries.¹ Moreover, it affects people of all ages, races, and socioeconomic backgrounds,¹ and accounts for a worldwide illness burden similar to that of breast cancer in women and lung cancer in men.²

Twenty percent of the epilepsies are medically intractable and mesial temporal lobe epilepsy related

to hippocampal sclerosis (MTLE-HS) is the most common form of surgically remediable epileptic syndrome.³ Anterior and mesial temporal lobectomy is a safe and effective treatment for medically intractable MTLE-HS.³ However, the literature regarding prognostic factors for the surgical outcome of MTLE-HS is often contradictory.⁴ Although prolonged febrile seizures, aberrant brain development, perinatal brain damage, and gene defects have been implicated, the pathophysiologic mechanisms involved in the development of MTLE-HS have not been completely elucidated.^{5,6} An association between MTLE-HS and polymorphism of the interleukin-1 α

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gene has been suggested,⁷ but not confirmed by another group.⁶ Variant alleles of the gene encoding the GABA(B) receptor 1 (G1465A)⁸ and the prodynorphin gene promoter⁹ have also been associated with temporal lobe epilepsy (TLE).

Cellular prion protein (PrP^c) is a glycoprotein anchored to the outer cell surface whose structure conversion is related to uncommon diseases, i.e., the spongiform encephalopathies.^{10,11} PrP^c is encoded by a single exon of a single gene that is highly expressed, predominantly in neurons, throughout adult life.¹¹ This protein binds copper and has antioxidant properties.^{12,13} In addition, PrP^c interacts with laminin and stress-inducible protein 1 (STI1), mediating neuronal cell adhesion, neurite extension and maintenance, and signal transduction.¹⁴⁻¹⁷

Ablation of the cellular prion protein gene (*PRNP*) enhances neuronal excitability of the hippocampus in vitro¹⁸⁻²⁰ and sensitivity to seizure in vivo,²¹ suggesting that total or partial modification of PrP^c function might be related to epilepsy.²²

We describe the presence of *PRNP* variant alleles in patients with clinically untreatable MTLE-HS and evaluate the association of these variant alleles with clinical, demographic, neurophysiologic, and neuroimaging data, as well as surgical outcome.

Methods. *Patients and controls.* DNA samples were collected from the peripheral blood cells of 100 consecutive patients surgically treated for MTLE-HS during their postoperative follow-up at the Ribeirão Preto Epilepsy Surgery Program between September 2000 and April 2002. The surgical treatment was indicated because seizures occurred at least monthly despite adequate use of anticonvulsants.

The inclusion criteria were as follows³: 1) seizure semiology consistent with MTLE, usually with epigastric, autonomic, or psychic auras, followed by behavioral arrest, progressive clouding of consciousness, orolimentary and manual automatisms, and autonomic phenomena; 2) anterior and mesial temporal interictal spikes; 3) no lesion other than atrophy and increased signal in hippocampal formation identified by MRI; 4) histopathologic examination compatible with hippocampal sclerosis; 5) presurgical investigation compatible with unilateral seizure onset; 6) absence of dual pathology that could be identified by available methods (clinical, electrophysiologic, and neuroimaging methods and histopathology); 7) at least 1 year of postoperative follow-up.

The exclusion criteria were as follows³: 1) focal neurologic abnormalities on physical examination; 2) generalized or extratemporal EEG spikes; 3) marked cognitive impairment on neuropsychological testing.

The control group included 180 healthy adults without a previous history of neurologic or psychiatric disease selected during the same period when we collected the cases. They were individuals accompanying patients from the internal medicine outpatient clinic of the same hospital as the patients. Similarly to the patients, all controls came from the same region of the state of São Paulo, were not related one to another, and had no personal or family signs or symptoms suggesting any type of spongiform encephalopathy. Patients and controls came from white (European Brazilians) and nonwhite (African Brazilians) backgrounds. There were no Amerindians or Asians in our sample.

Our research ethics committee approved the study, and informed consent was obtained from controls, patients, or persons legally responsible for them.

Clinical parameters and presurgical evaluation of patients. Clinical characteristics included sex, ethnicity, age at surgery, age at the onset of epilepsy (recurrent seizures), duration of epilepsy, a positive history for an initial precipitating insult (IPI), epilepsy duration until surgery, monthly complex partial seizure frequency impairing awareness in the year before surgery, and

side of surgery. A positive family history of epilepsy was defined when at least one first-degree relative reported at least two spontaneous seizures during his or her life.

Presurgical evaluation was performed by an experienced multidisciplinary team and included a detailed clinical history and neurologic examination, interictal and ictal video-EEG analysis, structural and functional imaging, psychiatric evaluation, neuropsychological testing, and, when appropriate, the intracarotid amobarbital test (Wada test) for memory and speech representation.²³ Detailed methods of neuroimaging and video-EEG analysis are available as supplementary data at www.neurology.org.

Surgical procedure and postoperative follow-up. The surgical approach was similar for all patients. A neurosurgeon experienced in surgery for epilepsy resected a maximum of 4 to 5 cm of the anterior lateral temporal lobe. Mesial resection included the amygdala and at least 1 to 3 cm of the anterior hippocampus (most commonly, 3.0 cm).

After surgery, patients were treated with the same antiepileptic drugs (AED) that they received before surgery and the dose was adjusted to reduce side effects and maintain the necessary serum levels. Patients were also instructed to maintain the medication dosage during the first 3 years after surgery even if they were free of seizures.

The outcome of surgery was defined on the basis of seizure status determined by outpatient clinical interviews. Patients were categorized as seizure free if they experienced no complex partial seizures or generalized tonic-clonic seizures after surgery. Nondisabling auras (Engel class Ib) were not considered seizures in our analysis. Patients who presented a single complex partial or a generalized tonic-clonic seizure clearly related to AED withdrawal (Engel class Id) or during the first month after surgery (perioperative period) but remained well controlled with adequate use of medication were considered seizure free.

A postoperative MRI was carried out 6 to 12 months after surgery. Surgical resection was considered satisfactory if it included both the head and body of the hippocampus, and unsatisfactory if no more than half of the hippocampus was resected.

Seizure outcome and postoperative MRI status were assessed by experienced epileptologists and radiologists who were blind to all clinical parameters, presurgical and surgical data, and the genetic analysis.

PRNP analysis. DNA was extracted from a 3 mL aliquot of whole blood using the Puregene Genomic DNA Isolation Kit (Gentra Systems). Primers (Life Technologies, S.P., Brazil) were designed to amplify two different overlapping fragments of the *PRNP* open reading frame (ORF) (Gene Bank: D-00015): fragment 1—nucleotides 77 to 497 (421 bp), forward: 5' ATGCTGGT-TCTCTTTGTG 3', reverse: 5' AACGGTCCTCATAGTCACTGC 3', and fragment 2—nucleotides 307 to 870 (563 bp), forward: 5' TCATGGTGGTGGCTGGGGTCA 3', reverse: 5' CGCCTCCCT-CAAGCTGGAAAA 3'. Amplification reactions were carried out using touchdown PCR. The PCR products were sequenced with the DYEnamic ET terminator sequencing kit (Amersham Pharmacia Biotech) according to manufacturer's instructions using an ABI Prism-377 apparatus (Perkin-Elmer). We also analyzed *PRNP* ORF using denaturing high-performance liquid chromatography (DHPLC) of PCR-amplified products using forward and reverse primers for fragment 1 (described above) and the forward (5' ATCATACATTTTCGGCAGT 3') and reverse (5' CTCCCTCAAGCT-GGAAAAAGA 3') primers for the second half of the *PRNP* ORF, nucleotides 463 to 867. A DNasep column (Transgenomic, CA) was used and the parameters of gradient and flow rate adjusted by the Wavemaker system control software (Transgenomic). PCR products from the complete coding region of *PRNP* obtained using the forward primer of fragment 1 and reverse primer of fragment 2 described above were digested with *PvuII*, *NspI*, and *BbvI* (Amersham Pharmacia Biotech and New England Biolabs) to confirm the sequence at codons 117, 129, and 171.²⁴⁻²⁶ In addition, due to the experimental difficulty in defining the exact position of the alteration within codon 117, a GC-rich DNA sequence,²⁷ we cloned (TA-cloning kit, Invitrogen) the PCR products of the complete coding region of *PRNP* from DNA from all patients and controls who presented an alteration at this codon.

Statistical analysis. Differences in sex and ethnicity distribution between cases and controls were analyzed by the two-tailed Fisher exact test. The proportions of cases and controls bearing a given *PRNP* genotype were ascertained. The magnitude of associ-

Table 1 Frequency of *PRNP* alleles in patients and controls

Residue position	Genotype	Exposed (%)		Crude		Adjusted§	
		Controls n = 180	Cases n = 100	OR (95% CI)	p Value*	OR (95% CI)	p Value
117	Ala/Ala	173 (96.1)	91 (91.0)	1.00		1.00†	
	Ala/Ala _{silent}	7 (3.9)	9 (9.0)	2.44 (0.88–6.78)	0.08	2.23 (0.78–6.38)	0.13
129	Met/Met	104 (57.8)	57 (57.0)	1.00		1.00‡	
	Met/Val	68 (37.8)	32 (32.0)	0.89 (0.50–1.46)	0.57	0.84 (0.49–1.44)	0.52
	Val/Val	8 (4.4)	11 (11.0)	2.51 (0.95–6.59)	0.06	2.35 (0.88–6.27)	0.08
171	Asn/Asn	180 (100.0)	77 (77.0)	1.00		—	—
	Asn/Ser	0 (0)	23 (23.0)	Undefined	<0.0001	NA	NA
Octarepeat	R12234	170 (94.4)	95 (95.0)	1.00		1.00	
	R12234/R1234	10 (5.6)	5 (5.0)	0.89 (0.28–2.69)	0.84	0.99 (0.33–3.03)	0.99
	R1234	0 (0)	0 (0)	—	—	—	—

* Binary Logistic Regression.

† Multiple logistic regression adjusted for gender, ethnicity, codon 129 distribution, and presence of the octarepeat deletion.

‡ Multiple logistic regression adjusted for gender, ethnicity, genotypes at position 117, and presence of octarepeat deletion.

§ Hosmer and Lemeshow goodness of fit, $p = 0.897$.

NA = non-applicable.

ation between *PRNP* alleles and the occurrence of MTLE-HS was measured by the OR and respective 95% CI. Crude and adjusted OR were estimated by unconditional logistic regression using the statistical program SPSS version 10.0 (Chicago, IL). To examine the independent effect of each genotype, OR were also derived after additional adjustment for codons 117 and 129. A p level < 0.01 was considered to be significant.

Association between a certain *PRNP* genotype and the clinical and presurgical evaluation data of patients was also determined. Analyses were done using the Mann-Whitney U test for continuous variables and the two-tailed Fisher exact test for categorical ones.

The Kaplan-Meier method was used to establish the cumulative probability of event-free survival. Differences between groups were determined by the log-rank test. The Cox proportional hazards model was applied to identify independent risk factors for having seizures after surgery. Only variables with $p \leq 0.25$ (by the log-rank test) were included in the final multiple Cox model.

For the statistical analysis of clinical and presurgical evaluation results the level of significance was set at $p < 0.05$. Our study fulfills the fundamental requisites when performing genetic investigations of complex multietiologic diseases.²⁸

Results. The sex and ethnic distribution were similar in patients and controls. Men represented 46% ($n = 84$) of the controls and 44% ($n = 44$) of the patients ($p = 0.7$). European Brazilians represented 86% ($n = 155$) of the controls and 88% ($n = 88$) of the patients ($p = 0.7$). The mean age of patients and controls was 36.2 (SD ± 10.7) and 36.5 (SD ± 10.5) years.

PRNP variant alleles were confirmed using three different experimental approaches: DNA sequencing, DHPLC, and endonuclease digestion (for codons 117, 129, and 171), and their frequencies in patients and in controls are shown in table 1. Two well-described *PRNP* polymorphic sites were present in our population. The single octapeptide repeat (24 bp) deletion (R12234)²⁹ was present in heterozygosity (R12234/R1234) in 5.0% of the patients and 5.6% of the controls ($p = 0.84$, OR = 0.89, 95% CI 0.28 to 2.69). The second polymorphic site was at codon 129 (Met→Val), which is located at the first PrP^c α -helix and is important for protein conformation.³⁰ The common polymorphism at

this codon appears to act as a genetic susceptibility factor in familial, sporadic, iatrogenic, and new variant Creutzfeldt-Jakob disease (CJD).³¹ In our population, heterozygous samples (Met/Val) at codon 129 had a nonsignificant decrease in OR for epilepsy ($p = 0.57$, OR = 0.89, 95% CI 0.50 to 1.46), whereas homozygous Val/Val showed a trend toward the opposite effect ($p = 0.06$, OR = 2.51, 95% CI 0.95 to 6.59), as compared to Met/Met individuals (see table 1). The octapeptide repeat deletion and codon 129 genotype frequencies in our controls were similar to those described in white populations.^{29,31}

Another polymorphic site at codon 117 where a GCA is altered to GCT without an amino acid change (Ala117Ala_{silent}) was present in heterozygosis in 7 (3.9%) of our controls and in 9 (9.0%) of the patients. This variant allele has been described in 5 to 10% of the normal white population,²⁶ close to the percentage observed in our cases and controls. A nonsignificant association with epilepsy was observed for those individuals carrying the Ala117Ala_{silent} allele ($p = 0.08$, OR = 2.44, 95% CI 0.88 to 6.78).

We additionally verified the presence of a variant allele in codon 171 (Asn→Ser), previously described in a familial form of atypical psychosis.³² The Asn/Ser heterozygous genotype was found in 23 (23.0%) of the patients and was absent in the control group ($p < 0.0001$) (see table 1), whereas the Ser/Ser genotype was not observed in our samples. This variant allele was first described as a rare polymorphism²⁴ and very recently Collinge's group was unable to detect it in a total of 944 tested individuals from East Asian, South Asian, Pacific, South American, and European populations.³³ Conversely, its frequency is around 5 to 8% in some African or African descendant populations (Cameroon and Jamaica).³⁴ Thus, the absence of this variant allele in our controls is in accordance with the literature and ethnicity is not a bias because more than 80% of both patients and controls are European Brazilians.

Table 2 Surgical outcome according to PRNP genotype

	All	Overall surgical outcome				Crude HR to remain with seizures	95% CI	<i>p</i> Value
<i>PRNP</i> variants	Patients n = 98 (%)	Seizure-free n = 85 (%)	Seizure n = 13 (%)	Cumulative % of seizure-free	<i>p</i> Value			
Position 117								
Ala/Ala	89 (90.8)	77 (90.6)	12 (92.3)	86.6		1.00		
Ala/Ala _{silent}	9 (9.2)	8 (9.4)	1 (7.7)	87.5	0.64	0.78	0.10–5.98	0.81
Position 129								
Met/Met	56 (57.1)	49 (57.6)	7 (53.8)	87.2		1.00		
Met/Val	31 (31.6)	27 (31.8)	4 (30.8)	86.3		1.04	0.30–3.56	0.95
Val/Val	11 (11.2)	9 (10.6)	2 (15.4)	79.5	0.86	1.51	0.31–7.29	0.60
Position 171								
Asn/Asn	76 (77.6)	70 (82.4)	6 (46.2)	91.8		1.00		
Asn/Ser	22 (22.4)	15 (17.6)	7 (53.8)	68.2	0.005	4.12	1.38–12.77	0.01
Octarepeat								
R12234	93 (94.9)	80 (94.1)	13 (100.0)	86.2		1.00		
R12234/R1234	5 (5.1)	5 (5.9)	0 (0.0)	100.0	0.41	NA	—	—

The distribution of Asn/Ser heterozygous alleles in patients was not changed by genotypes at positions 117 or 129 or octapeptide repeat deletion as verified by stratified analysis, suggesting that this variant allele might be independently associated with epilepsy (data not shown).

OR were also derived after additionally controlling for the effect of genotypes at positions 117 and 129, the presence of the octarepeat deletion, sex, and ethnicity (see table 1). Neither the Met/Val genotype at position 129 nor the presence of the octarepeat deletion was found to be associated with MTLE-HS, whereas a trend for a higher frequency of Val/Val genotypes at position 129 was observed in the patients ($p = 0.08$, adjusted OR 2.35, 95% CI 0.88 to 6.27). Individuals carrying the silent variant allele Ala/Ala_{silent} at position 117 also disclosed a nonsignificantly increased OR for epilepsy ($p = 0.13$, adjusted OR 2.23, 95% CI 0.78 to 6.38). Independent effects of genotypes at position 171 could not be evaluated.

No other PRNP variant alleles besides those described here were found in cases or controls. All genotype distributions studied were in good agreement with Hardy-Weinberg equilibrium (data not shown).

Because all our patients were surgically treated we were able to perform histopathologic examinations of resected material and no spongiform degeneration was found in brain tissue.

No differences were found between PRNP variant alleles and the wild type genotype ($p \geq 0.15$) with respect to the clinical, demographic, and presurgical data investigated. There was a nonsignificant increased OR ($p = 0.44$, OR = 1.67, 95% CI 0.6 to 4.2) for positive family history of epilepsy in those patients with the variant allele Asn171Ser. For more details, see supplementary data at www.neurology.org.

The postoperative seizure outcome according to patient PRNP genotype is shown in tables 2 and 3; data related to patient characteristics can be found in supplementary tables E-1 and E-2 (available at www.neurology.org). Two patients who presented surgical complications that could

affect seizure outcome (meningitis and hemorrhagic stroke) were excluded from this analysis. The mean time of postoperative patient follow-up was 3.1 years (SD ± 1.3 , range 1.0 to 7.5 years). Of the 98 patients, 85 (86.7%) remain seizure free after surgery. This is in agreement with more recent data suggesting that, with careful selection of patients, a good seizure outcome may be achieved in approximately 90% of cases.^{4,35}

The seizure outcome was not correlated with any preoperative clinical, neurophysiologic, or neuroimaging data (see table 3 and supplementary data [www.neurology.org]). Variant alleles at positions 117 ($p = 0.64$) and 129 ($p = 0.86$) and the octarepeat deletion ($p = 0.41$) were also not correlated with surgical outcome (see table 2). However, the hazard ratio to continue to have seizures after temporal lobectomy in those patients carrying the Asn171Ser variant allele was more than four times higher than in those with the normal allele ($p = 0.01$, 95% CI 1.38 to 12.77) (see table 2). After an 18-month follow-up, the cumulative probability of being seizure free was 91.8% for patients with normal alleles at codon 171 (Asn171Asn) and only 68.2% for those with the Asn171Ser variant allele in heterozygosity ($p = 0.005$) (figure). After adjustment for ethnicity, history of IPI, presurgical MRI findings, interictal spikes, and age at epilepsy onset (see table 3), the risk for surgical failure in those patients with the variant allele at codon 171 remained high ($p = 0.005$, adjusted HR = 5.35, 95% CI 1.65 to 17.33). Positive history of IPI, bilateral findings on the presurgical MRI, presence of bilateral interictal spikes, and early age at epilepsy onset were not independently associated with a worse surgical outcome in our sample. Independent effects of ethnicity could not be evaluated.

One patient showing the variant allele at codon 171 had a generalized seizure in the first week after surgery, and became well during the next 2-year follow-up. Among the 15 seizure-free patients showing the abnormal variant allele at codon 171, one had a complex partial seizure related to AED withdrawal (Engel Id), in comparison with 8 of the

Table 3 Independent effect of the PRNP variant allele on the surgical outcome

Variable	Crude HR	95% CI	Adjusted HR*	95% CI	p Value
History of IPI					
No	1.00		1.00		
Yes	1.93	0.65–5.73	0.48	0.15–1.57	0.23
MRI findings					
Unilateral	1.00		1.00		
Bilateral	2.53	0.53–11.4	1.75	0.36–8.40	0.48
Interictal spikes†					
Unilateral	1.00		1.00		
Bilateral	2.45	0.67–9.11	1.58	0.38–6.45	0.52
Age at epilepsy onset					
After the first decade	1.00		1.00		
First decade	2.36	0.72–7.65	2.03	0.50–8.36	0.32
Position 171					
Asn/Asn	1.00		1.00		
Asn/Ser	4.12	1.38–12.77	5.35	1.65–17.33	0.005

* Hazard-ratio to continue to have with seizure after surgery adjusted for ethnicity, history of IPI, pre-surgical MRI findings, IS, and age at epilepsy onset.

† Interictal spikes were not quantified in three patients.

IPI = initial precipitating insult.

70 showing the normal allele ($p = 1.0$). Patients with AED withdrawal became well controlled with an adequate AED scheme during a mean follow-up of 3.4 years (range 1.7 to 5.5 years).

All patients who underwent semi-invasive investigation with foramen ovale electrodes had the Asn171Asn genotype.

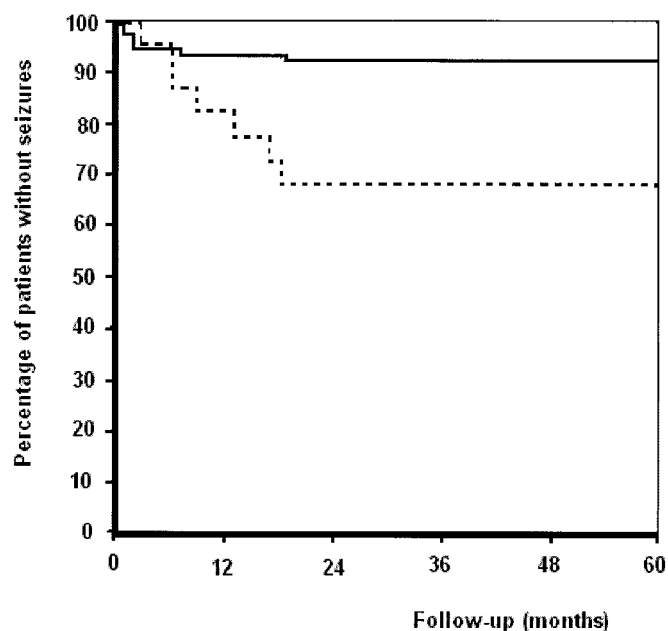


Figure. Kaplan-Meier event-free survival curves comparing the cumulative percentages of patients who were free of seizures according to the genotype at codon 171. There were more seizure-free patients with the Asn171Asn allele than patients with the Asn171Ser variant ($p = 0.005$ by log-rank test). Genotype: — = Asn171Asn; - - - = Asn171Ser.

The eight cases with bilateral MRI findings presented symmetrically increased signals in the hippocampus but asymmetric hippocampal atrophy, which was coincident with the inferred epileptogenic zone (EZ). One patient with bilateral hippocampal atrophy was excluded from the outcome analysis due to postoperative meningitis. Finally, among the seven patients with bilateral hippocampal atrophy, all the seizure-free patients ($n = 5$) presented the Asn171Asn genotype; the others who persisted with seizures ($n = 2$) had the Asn171Ser variant allele ($p = 0.048$).

Discussion. Using a candidate gene approach, we found that the PRNP variant Asn171Ser is highly associated with medically untreatable MTLE-HS and independently influences the postsurgical seizure outcome of patients.

The association between PRNP and epilepsy and especially with MTLE-HS is plausible and supported by previous in vivo and in vitro experimental data. Mice devoid of PRNP presented a higher sensitivity to seizures chemically induced by different agents in four different protocols.²¹ This hyperexcitability can be related or attributed to the interaction of at least five distinct abnormalities reported in those animals: 1) disrupted Ca^{+2} -activated K^{+} currents, with loss of I_{HAP} conductance in the hippocampus¹⁸; 2) abnormal GABA-A inhibition in the hippocampus¹⁹; 3) mossy fiber reorganization in the hippocampus³⁶; 4) higher levels of oxidative stress in the neocortex and hippocampus¹³; and 5) changes in ectonucleotidases in both hippocampus and neocortex.³⁷ Furthermore, postnatal PrP^c knockout mice showed a significant reduction of after hyperpolarization potentials in the hippocampus.²⁰ These findings suggest that altered PrP^c functions could contribute to modifications of the intrinsic physiology of neurons and the neural

network related to mechanisms of epileptogenesis, particularly in the mesial temporal lobe structures.²² The exact role of *PRNP* variant alleles in PrP^c functions and in MTLE-HS remains to be determined. Position 171 is close to the site where PrP^c interacts with laminin, mediating neuritogenesis,¹⁵ which is closely related to neuroplasticity and epileptogenesis.²²

It would be important to establish why modifications in an abundantly expressed neuronal protein would affect focal epileptogenicity. A possible explanation could be that changes in PrP^c function (in humans) or the absence of PrP^c (in knockout mice) do not cause epilepsy but induce a proepileptogenic state that, in association with other inborn (genetic or developmental) or acquired (lesion) abnormalities (including a surgical scar), predisposes to temporal lobe epileptogenesis. It was surprising that the *PRNP* variant allele Asn171Ser does not correlate with presurgical variables; however, this finding is in agreement with the lack of association between those already studied presurgical variables and surgical outcome. Association between variant alleles, seizure semiology, and ictal electrophysiology patterns in MTLE-HS or even in other epileptic syndromes is an important point to be investigated in the future.

Although nonsignificant, the trend toward an association between epilepsy and the variant allele Val129Val is still interesting. The critical localization of amino acid 129 at the first PrP^c α -helix renders it important for protein conformation.³⁰ We speculate that the amino acid composition at this position might be directly related to protein function or interfere with other PrP^c domains with biologic activity.

The *PRNP* variant allele at codon 171 was found in many but not all patients with epilepsy. In addition, despite the association of the Asn171Ser variant allele with surgical failure, many patients who presented the variant allele became seizure free. These findings suggest that *PRNP* contributes only partially to the penetrance of MTLE-HS and its surgical outcome, possibly providing a genetic liability for developing the condition. However, only those individuals whose liability crosses the threshold would actually be at risk. In fact, it is well known that epilepsies in general are caused by the interaction of genetic and environmental factors. In these cases, as for other non-mendelian diseases, the combination of susceptibility alleles across the genome that is necessary for disease development provides only a risk of developing the disorder.^{38,39} The lack of association between familial history of epilepsy or seizures and the variant alleles also suggests that a complex pattern of inheritance is involved.⁸

The *PRNP* variant allele Asn171Ser is highly prevalent in patients with untreatable MTLE-HS and influences their surgical outcome, suggesting that *PRNP* modifications may play a role in the process of epileptogenesis of the most common type of surgically remediable epileptic syndrome.³ If con-

firmed by studies in independent patient population samples, determination of *PRNP* sequences could improve the diagnosis and identification of high-risk groups, provide a basis for genetic counseling, and possibly lead to new therapeutic targets for the prevention and treatment of MTLE-HS.

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Randomized, concentration-controlled trial of topiramate in refractory focal epilepsy

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Abstract—Objective: To establish the concentration response of topiramate in patients with refractory focal epilepsy. **Methods:** Sixty-five patients with more than eight seizures during an 8-week baseline were randomized to three prespecified plasma levels (low, 6 $\mu\text{mol/L}$ [2 mg/L]; medium, 31 $\mu\text{mol/L}$ [10.5 mg/L]; and high, 56 $\mu\text{mol/L}$ [19 mg/L]). Topiramate treatment was titrated to one of the prespecified plasma levels during an 8-week titration period, followed by a 12-week observation period. **Results:** The overall median (25th to 75th percentile) reduction in seizures during the observation compared with baseline was 50% (9.5 to 90%). In the individual groups, the median reduction was as follows: low, 39% (13 to 70%); medium, 85% (41 to 96%); and high, 39% (2.0 to 81%). The primary outcome of the trial was the comparison of seizure reduction (Mann–Whitney *U* test) between the low and the medium group ($p = 0.03$). Comparisons between the other groups were as follows: medium vs high ($p = 0.05$) and low vs high ($p = 0.81$). Psychiatric adverse events and adverse events related to the CNS were the most frequently encountered. Most adverse events showed concentration response, particularly between low and medium levels. **Conclusions:** Patients assigned to the medium plasma level (31 $\mu\text{mol/L}$ [10.5 mg/L]) had the best seizure outcome. Patients in the medium and high groups experienced more adverse events than patients in the low group. Optimal treatment response is thus most likely found at plasma concentrations higher than 6 $\mu\text{mol/L}$ (2 mg/L), but no further increase in efficacy seems to occur at concentrations above 31 $\mu\text{mol/L}$ (10.5 mg/L).

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The effect of the new antiepileptic drug topiramate has been well established in several add-on trials in patients experiencing partial seizures.^{1–5} Dose response has been established from clinical trials, all employing a similar design.^{1–5} Response to topiramate was larger with 400 to 600 mg/day compared with 200 mg/day, but there was no further increase in response in patients treated with ≥ 600 mg/day.^{1–5}

The relationship between the plasma concentration of topiramate and efficacy has not been firmly established, and none of the studies mentioned above was concentration controlled. Indeed, problems with the identification of a relationship between plasma concentration and response may stem from choice of an inappropriate study design. To gain valid concentration–response information, Gram et al⁶ argued that

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